

Short-term wind power forecasting based on meteorological feature extraction and optimization strategy

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ABSTRACT

Accurate wind power forecasting is a vital factor in day-ahead dispatch and increasing the level of penetration of renewable energy. The feature extraction of meteorological factors related to wind power output is a big challenge to improve forecasting accuracy, and selecting key meteorological factors based on experience will decrease the prediction accuracy. Therefore, a day-ahead wind power combined forecasting approach is innovatively proposed through key meteorological factors selection, data decomposition and reconstruction, combined forecasting model generation, and optimization strategy. Correlated variables namely variational mode decomposition and weighted permutation entropy (VMD-WPE) decomposed historical wind power and key meteorological factors are used as the inputs. A forecasting model based on convolutional neural network (CNN) and long short-term memory network (LSTM) is used to forecast future wind power. Four optimizers with different optimization performances are used to find the best parameters of the forecasting model to obtain accurate prediction results. Multiple comparative experiments from regional wind farms in Ningxia and Jilin of China are utilized as case studies to evaluate the effectiveness of the proposed model. Results show that the proposed approach outperforms other benchmark prediction models, taking into account multiple-error metrics including error metrics, accuracy rate, and improvement percentages.

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1. Introduction

With the reduction of global fossil energy, the development of renewable energy has gradually attracted the attention of governments around the world. As a typical representative of renewable resources, wind energy has been widely used due to its technological maturity and easy accessibility. In recent years, developers have made full use of inland and coastal wind energy resources, resulting in a continuous increase in installed wind power capacity in China [1,2]. Fig. 1 shows the new and cumulative wind power installed capacity from 2010 to 2019 in China, which indicated that wind power utilization has achieved great success. Affected by wind speed, wind power shows strong uncertainty, resulting in large fluctuations in wind power output, which brings big challenges to the integration of wind power [3]. Therefore, effective short-term wind power can reduce the adverse effects of wind power intermittence and fluctuation, and improve the power system to accommodate wind power.

Wind power forecasting can be divided into multi-time horizons according to the application scenarios of power system dispatching and planning [4,5], which are ultra-short-term, short-term, medium-term, and long-term. In the ultra-short-term, the time horizon ranging from a few minutes to several hours (0–4 h), which can help power system operators to control wind power and real-time dispatching [6]. The prediction horizon of the short-term is from hours to days (hours-days), the main task is to make an intraday or day-ahead dispatching planing (economic dispatch) [7]. In the medium-term, the prediction horizon is from days to weeks (days-weeks), and it can help the manager to coordinate wind power and conventional unit commitment and maintenance dispatching [8]. In the long-term, the conventional power plants and wind power transmission lines are managed [9]. When it comes to day-ahead dispatching of thermal power units, short-term wind power prediction is the primary consideration.

In terms of ultra-short-term power prediction, statistical methods, machine learning, and combined methods are often used. Statistical models are commonly used to forecast wind power time series, which including Autoregressive moving average (ARMA), Autoregressive integrated moving average (ARIMA), and

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Nomenclature

NWP	Numerical weather prediction
CNN	Convolutional neural network
LSTM	Long short-term memory network
WPE	Weighted permutation entropy
WPF	Wind power forecasting
ARMA	Autoregressive moving average model
ARIMA	Autoregressive integrated moving average
GARCH	Generalized autoregressive conditional heteroscedasticity
ARX	An autoregressive with exogenous variable
WT	Wavelet transform
ANNs	Artificial neural networks
RBF	Radial basis function neural network
DBN	Deep belief network
CNN	Convolutional neural network
SVM	Support vector machine
GRNN	Generalized regression neural network
ENN	Elman neural networks
VMD	Variational mode decomposition
AR	Autoregressive
GP	Gaussian process
CF	Covariance function
ELM	Extreme learning machine
FFBP	Feed-forward back propagation
GSA	Gravitational search algorithm
ICS	Improved cuckoo search
LSTM	Long short-term memory
p.u.	Per unit
Symbols and notation	
$p(t)$	Wind power
u_k	Wind power unilateral spectrum
$\delta(t)$	Dirac distribution
$e^{-j\omega_k t}$	Center frequency
λ	The lagrange multiplier
α	Balance parameter of the data fidelity constraint

$\hat{p}(\omega)$	Fourier transform of wind power
$\hat{m}_i(\omega)$	The fourier transform of mode
λ_t	The fourier transform of lagrange multiplier
m	The embedding dimension
$H(m)$	The entropy value
ω_j	The weight value
$\overline{\text{mode}_j^m}$	The average value
Mfi	The selected meteorological factors
Mode i	The decomposition wind power
$\varphi(\cdot)$	The activation function
ω	The vector of weighting factors in kernels
n	The number of kernels
b	The vector of bias
$\otimes$	The convolution operation
σ	The sigmoid activation function
$\omega_{fv,p}$	The weight matrices of the forgetting gate
b_f	The bias vector
f_t	The output results of the forgetting gate
$\omega_{gv,p}$	The weight matrices of the input gate
$\omega_{gh,p}$	The weight matrices of the input node
b_g	The corresponding bias vector
$h(t)$	The output results of the implicit state
Std.dev	The standard deviation
Max	Maximum
Min	Minimum
$NMAE$	Normalized mean absolute error
$NRMSE$	Normalized root mean square error
C_R	The forecasting accuracy
Δ_{NMAE}	The improvement percentages of $NMAE$
Δ_{NRMSE}	The improvement percentages of $NRMSE$
$NMAE_{\text{benchmarkmodel}}$	The $NMAE$ of benchmark
$NMAE_{\text{proposedmodel}}$	The $NMAE$ of proposed model
$NRMSE_{\text{benchmarkmodel}}$	The $NRMSE$ of benchmark
$NRMSE_{\text{proposedmodel}}$	The $NRMSE$ of proposed model
$p_{i,t}^{\text{pre}}$	The prediction value at time t
$p_{i,t}^{\text{real}}$	The actual wind power at time t

Generalized autoregressive conditional heteroscedasticity (GARCH). An autoregressive with exogenous variable (ARX) model is developed, the study demonstrates that ARX can improve accuracy compared with conventional autoregressive (AR) models [10]. To get the best most suitable configuration of the Kalman model, the ARIMA is employed [11]. Further, data decomposition techniques such as Wavelet Transform (WT) incorporate ARIMA into a new model to forecast short-term wind speed [12]. In Ref. [13], a combined forecasting model based on ARMA-GARCH is proposed, in which the result shows that the proposed method can capture the fluctuation characteristics of wind speed. All these models have a great contribution to ultra-short-term wind power. However, in the implementation process, once the model parameters are selected, the time-ahead may not applicable.

Machine learning methods that are popularly applied to wind power/speed forecasting can be further classified into two groups: artificial neural networks (ANNs) and deep learning. The artificial neural network is used for wind power or wind speed prediction, and the accuracy will decrease with the time horizon increases. In terms of wind speed forecasting, ANNs (e.g., two-layers ANN [14], radial basis function neural network (RBFNN) [15], improved RBFNN [16], an ensemble of mixture density ANN [17], deep ANN [18]) are used to forecast wind speed, simulation results show that

most ANN-based models can obtain high accuracy forecasting value compared to the methods without artificial intelligence. The other kind of model is deep learning, which including deep belief network (DBN) [19], convolutional neural network (CNN), and long short-term memory (LSTM) [20]. These models have powerful nonlinear mapping functions, which make deep learning methods have better performance. DBN has an adaptive learning ability that can capture the characteristics of wind speed changes and effectively improve the accuracy of prediction [21]. CNN has the advantage of extracting nonlinear features from input samples. In Ref. [22], a data feature extraction model based on a three-dimensional convolutional neural network (3D-CNN) is proposed, which can extract spatiotemporal features from NWP data. LSTM is more suitable for ultra-short-term wind power time prediction due to its special structure design. In Ref. [23], a deep learning prediction method based on LSTM is proposed, which including data decomposition, feature selection, and parameters optimization, and a promising result is obtained.

At present, the combined methods have a strong adaptive ability in coping with non-stationary signals, and more and more scholars use these methods in wind power forecasting [24–26]. The combined methods include data decomposition technique, data reconstruction technique, forecasting model, and parameter

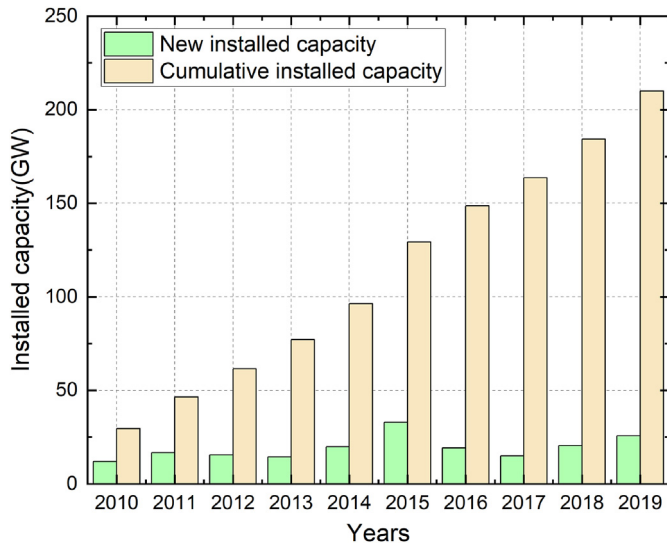


Fig. 1. New and cumulative wind power installed capacity from 2010 to 2019 in China.

optimization method. The wind power/wind speed time series is decomposed into sub-sequences in different frequency domains, which can characterize the time domain wind power as a frequency domain signal, and each signal is presented by a specified prediction model [27,28]. However, when the number of wind power decomposition increases, the training time will increase. To solve this problem, the entropy theory is developed to evaluate the complexity of decomposition information to reconstruct a new sequence [29]. In terms of choosing the forecasting model, artificial neural networks and machine learning are used as a predictor [30]. Parameter optimization can improve model reliability and prediction accuracy to a certain extent. The initial weights and thresholds of the back propagation neural networks (BPNN) NN or RBFNN, the weights and bias of the extreme learning machine (ELM), the smoothing factor of generalized regression neural network (GRNN), and the learning rate of Elman neural networks (ENN), can be optimized by using different meta-heuristic algorithms [31–33]. These methods can improve the reliability and robustness of the forecasting model, however, few works have been done in optimizing the model based on meteorological factors and decomposition wind power as the inputs.

In terms of short-term power prediction, the prediction method based on a neural network is mainly used [34]. The best meteorological input variable is selected as the model input by searching for the nonlinear mapping relationship between the numerical weather prediction (NWP) meteorological factors and wind power. In previous literature, meteorological factors such as wind speed, wind direction, air pressure, and temperature are often selected based on experience as key meteorological factors, which may result in low accuracy of short-term wind power forecasting and unable to meet the day-ahead scheduling.

Based on the above analysis, a conclusion can be obtained. All the above methods can be summarized into two categories according to the characteristics of the input variables. The first category is wind power time series forecasting. The historical power is used as the input variable at time t to predict wind power at time $t+1$ [35]. This type of method is suitable for very-short-term wind power forecasting, and the accuracy of short-term wind power forecasting needs to be improved. The other type mainly uses a neural network-based forecasting method. This method establishes a prediction model by finding the nonlinear mapping relationship

between NWP meteorological factors and wind power, which can obtain a better accuracy for the flat wind power output. When it comes to the decomposed power and meteorological factors as the sample data training model, to the authors' knowledge, researchers have paid little attention to this topic. Motivated by this, a combined forecasting framework including key weather factor selection, wind power data decomposition, and reconstruction, model input feature extraction, parameter optimization is developed, and a relatively reliable result can be obtained. The main contributions are as follows:

- 1) Selecting key meteorological factors: Meteorological factors can provide valuable information as input variables of the prediction model, and analyzing the correlation between NWP data and wind power is essential to improve data quality. The Pearson correlation coefficient method is used to calculate the correlation between NWP meteorological variables and wind power, and the best input meteorological variables are selected.
- 2) Data decomposition and reconstruction: The variational modal decomposition method is used to decompose wind power into several modes with different characteristics, which can solve the problem of the autocorrelation of wind power time series prediction. Furthermore, the weighted permutation entropy is used to reconstruct the modes into a new mode with high frequency, intermediate frequency, and low frequency, thereby reducing the redundancy of input variables.
- 3) Input feature extraction of forecasting model: CNN is used to extract key factors and reconstruction new modes through convolution, kernel, and pooling operations, thereby overcoming the shortcomings of traditional neural networks that lack useful information extraction. The variables extracted by CNN are input to the LSTM prediction model. In this step, different optimizers are used to tune the parameters of the prediction model, and the forecasting values based on different optimizers can be obtained.
- 4) Aggregate technique: Using the aggregation technique to re-add the prediction results obtained from different optimizers, and finally obtain the day-ahead wind power forecasting.

The rest of the paper is organized as follows. In Section 2, the problem statement and methodology are provided. Section 3 describes the data analysis and performance criteria. Results and analysis are described in Section 4. Discussion is given in section 5. Conclusions and future works are given in Section 6.

2. Problem statement and methodology

In previous studies, meteorological factors (e.g., wind speed, wind direction, air temperature, and air pressure) are often selected as the input of the forecasting model based on experience, which can be assumed to improve prediction accuracy. However, this approach may not quantitatively evaluate the contribution of meteorological factors to wind power forecasting. Therefore, a day-ahead wind power combined forecasting framework is proposed to improve prediction accuracy. The overall framework of the proposed model is shown in Fig. 2. Furthermore, the following problem needs to be solved:

1. Selecting the best meteorological factor: It is necessary to develop a method that can effectively characterize the correlation between NWP and historical wind power data. This method should contain sufficient information about the contribution of uncertain meteorological factors to wind power.
2. Decomposing the wind power from the time domain to the frequency domain: In the decomposition procedure, parameter

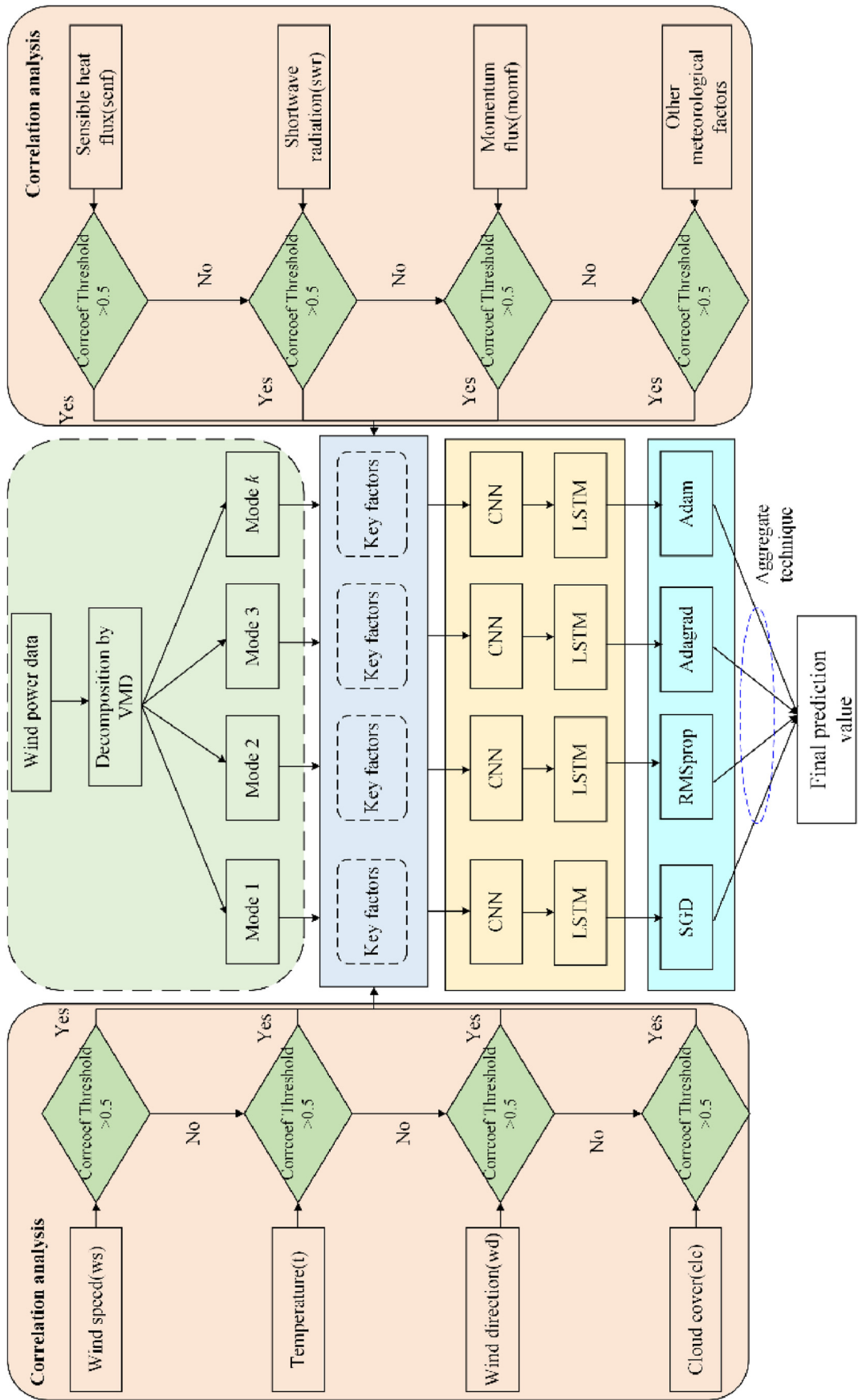


Fig. 2. The overall framework of the proposed model.

estimation should be avoided, and the decomposition method should be able to characterize the trend of wind power fluctuation. Further, the data reconstruction method can obtain complete and useful decomposition data.

3. Forecasting model generation: The combined forecasting model has become the mainstream method, and it is necessary to explore a forecasting framework that can be used for spatio-temporal data mining, which can avoid the appearance of large wind power forecasting errors.
4. Developing a new optimization strategy: It is necessary to develop an optimization strategy to meet the requirements of different input variables for the forecasting model, and the strategy can tune the parameters.

2.1. Wind power decomposition

In terms of the data decomposition method, the variational mode decomposition (VMD) algorithm decomposes the wind power time series into several modes by finding the optimal solution of the constrained variational model. In the decomposition procedure, the center frequency and bandwidth of each component of the mode are updated alternately and iteratively. Finally, through frequency band-based signal adaptive decomposition, K narrow-band modes can be obtained. The procedure of VMD is described as follows:

For each mode $m_k(t)$, the wind power unilateral spectrum u_k can be obtained by the Hilbert transform, which is expressed as:

$$u_k = \left[\delta(t) + \frac{j}{\pi t} \right] * m_k(t) \quad (1)$$

where $\delta(t)$ denotes Dirac distribution and $*$ denotes convolution.

The unilateral spectrum is moved to the baseband, and multiply the center frequency $e^{-j\omega_k t}$ by the analysis signal of each mode, which can be rewritten as:

$$u_k = \left\{ \left[\delta(t) + \frac{j}{\pi t} \right] * m_k(t) \right\} \cdot e^{-j\omega_k t} \quad (2)$$

The bandwidth of each mode function is obtained by Gaussian smoothing, and the sum of the modes is the decomposed wind power $p(t)$, the resulting constrained variational problem is formulated as:

$$\begin{cases} \min_{u_k, \omega_k} \left\{ \sum_{k=1}^K \left\| \partial_t \left\{ \left[\delta(t) + \frac{j}{\pi t} \right] * u_k(t) \right\} \cdot e^{-j\omega_k t} \right\|_2^2 \right\} \\ \text{s.t.} \sum_{k=1}^K u_k(t) = p(t) \end{cases} \quad (3)$$

where s.t. is short for subject to.

The Lagrange multiplier λ is introduced to transform the constrained variational problem into an unconstrained variational problem, which is expressed as follows:

$$\begin{aligned} L(\{m_k\}, \{\omega_k\}, \lambda) = & \alpha \sum_{k=1}^K \left\| \partial_t \left\{ \left[\delta(t) + \frac{j}{\pi t} \right] * m_k(t) \right\} \cdot e^{-j\omega_k t} \right\|_2^2 \\ & + \left\| f(t) - \sum_k m_k(t) \right\|_2^2 + \langle \lambda(t), f(t) - \sum_k m_k(t) \rangle \end{aligned} \quad (4)$$

where λ is the Lagrangian multiplier, and α is the balance parameter of the data fidelity constraint.

The saddle point of the extended Lagrange expression is obtained by alternately updating m_k and ω_k . Finally, the frequency domain and center frequency update of each mode is given as follows:

$$\begin{cases} \hat{m}_k^{n+1} = \frac{\hat{p}(\omega) - \sum_{i \neq k} \hat{m}_i(\omega) + \frac{\hat{\lambda}(\omega)}{2}}{1 + 2\alpha(\omega - \omega_k)} \hat{\omega}_k^{n+1} = \frac{\int_0^\infty \omega |\hat{m}_k(\omega)|^2 d\omega}{\int_0^\infty |\hat{u}_k(\omega)|^2 d\omega} \end{cases} \quad (5)$$

where n denotes the number of iterations, $\hat{p}(\omega)$, $\hat{m}_i(\omega)$, and $\hat{\lambda}(\omega)$ implicate the Fourier transform of $p(t)$, $m_i(t)$, and λ_t , respectively.

2.2. Wind power reconstruction based on weighted permutation entropy

Weighted permutation entropy (WPE) is used to evaluate the complexity of wind power [36]. For the decomposed wind power $\{\text{Mode}_i^m\}_{i=1}^N$, where N is the size of wind power, and generate a new $\text{Mode}_j^m = \{\text{Mode}_j, \text{Mode}_{j+1}, \dots, \text{Mode}_{j+m-1}\}$, where m is the embedding dimension. Each new Mode_j^m is sorted in ascending order based on the entropy value, and PE is expressed as follows:

$$H(m) = - \sum_{i: \pi_i^m \in \Pi} p(\pi_i^m) \log p(\pi_i^m) \quad (6)$$

where Π denotes the $\{\pi_i^m\}_{i=1}^{m!}$, and $p(\pi_i^m)$ is given as follows:

$$p(\pi_i^m) = \frac{\{j|j = 1, \dots, N - m + 1; X_j^m \text{ has type } \pi_i^m\}}{N - m + 1} \quad (7)$$

In formula (7), the value range of PE is $[0, \log m!]$. Recently, weighted permutation entropy (WPE) is used to extract amplitude information of Mode_j^m , and the Mode_j^m is given a weight ω_j , and the weighted relative frequencies can be written as:

$$p_\omega(\pi_i^m) = \frac{\sum_{j: X_j^m \text{ has type } \pi_i^m} \omega_j}{\sum_{j=1}^{N-m+1} \omega_j} \quad (8)$$

where $\sum_i p_\omega(\pi_i^m) = 1$.

The weight value ω_j of the Mode_j^m is simplified as follows:

$$\omega_j = \frac{1}{m} \sum_{k=1}^m [\text{Mode}_{j+k-1} - \overline{\text{Mode}_j^m}]^2 \quad (9)$$

The average value $\overline{\text{Mode}_j^m}$ is formulated as follows:

$$\overline{\text{IMF}_j^m} = \frac{1}{m} \sum_{k=1}^m \text{imf}_{j+k-1} \quad (10)$$

Then the WPE is described as:

$$H_\omega(m) = - \sum_{i: \pi_i^m \in \Pi} p_\omega(\pi_i^m) \log p_\omega(\pi_i^m) \quad (11)$$

When only $\omega_1 = \omega_2 = \dots = \omega_{N-m+1}$, WPE reduces to PE.

2.3. Convolutional neural network

A convolutional neural network (CNN) is used to extract spatial features of input data that are strongly related to day-ahead wind power forecasting. To improve the effectiveness of complex data modeling, the CNN algorithm usually sets three mapping layers including convolutional layer, pooling layer, and fully connected layer. The convolutional layer is used to find local relationships in the input, while the pooling layer gradually reduces the dimensionality relative to the target variable, the fully connected layer is used to forecast the target variable based on the features in the input variable. Moreover, CNN uses the local connection, weight sharing, and spatial or temporal correlation sampling methods to obtain better features. Compared with conventional models with the same layers and neurons, CNN can be trained very efficiently.

The input of CNN consists of feature vectors on consecutive M time steps. Assume that X is the feature vector of the t time step, the formula is defined as:

$$X_t = [Mf_{1t}, Mf_{2t}, Mf_{3t}, \dots, Mf_{nt}, Mode_{1t}, Mode_{2t}, \dots, Mode_{kt}] \quad (12)$$

where $Mf_i (i = 1, 2, \dots, n)$ is the selected meteorological factors, $Mode_i (i = 1, 2, \dots, k)$ is the reconstruction wind power.

In the convolutional layer, a convolution calculation is used to fuse two sets of information, and the convolutional layer of CNN transforms the sample input data, and the convolution operation is defined as:

$$feature = \varphi(\omega^n \otimes x + b^n) \quad (13)$$

where $\varphi(\cdot)$ is the activation function. ω , n , x and b denote the vector of weighting factors in kernels, the number of kernels, the vector of input series, and the vector of bias, respectively. The symbol $\otimes$ is the convolution operation.

The weight update of CNN uses the error back propagation algorithm to continuously adjust the network connection weight to minimize the forecasting error, in this way, the convolutional layer and the pooling layer can learn the optimal parameter matrix.

2.4. Long-short term memory wind power forecasting

In the field of wind power forecasting, to solve the explosion gradient problem of traditional neural networks, the long-short term memory (LSTM) model is proposed. The LSTM introduces a specially designed LSTM unit, which can control the flow of hidden state information of wind power from one time step to another step. The structure of the LSTM unit is shown in Fig. 3. Assume that x_t represents the input vector at time t , h_{t-1} represents the last implicit state, and s_t represents the vector stored in the external storage unit, which can transmit information between different time steps. The procedure of LSTM implementation requires three gates: forget gate, input gate, and output gate.

A forget gate element i_t is shown as below:

$$f_t = \sigma(\omega_{fv,p}x_t + \omega_{fh,p}h_{t-1} + b_f) \quad (14)$$

where σ is the sigmoid activation function, $\omega_{fv,p}$ and $\omega_{fh,p}$ are the weight matrices of the forgetting gate, b_f is the bias vector, f_t is the output results of the forgetting gate.

The input gate and output gate are as follows:

$$\begin{cases} g_t = \varphi(\omega_{gv,p}x_t + \omega_{gh,p}h_{t-1} + b_g) \\ s_t = s_{t-1} \odot f_t + g_t \odot i_t \end{cases} \quad (15)$$

where $\omega_{gv,p}$ and $\omega_{gh,p}$ are the weight matrices of the input gate and input node, and b_g is the corresponding bias vector, $\varphi(\cdot)$ is the tanh activation function, the $s_{t-1} \odot f_t$ is to determine how much information will be forgotten from c_t ; $g_t \odot i_t$ determines how much information is added to the new unit state s_t . $\odot$ is Hadamard product.

Using the output gate $o(t)$ to calculate $h(t)$

$$h(t) = \varphi(s(t)) \odot o(t) \quad (16)$$

where $h(t)$ is the output results of the implicit state.

2.5. The method for wind power forecasting

A framework of wind power is proposed to improve the accuracy of day-ahead wind power prediction, which includes key meteorological feature selection, data decomposition and reconstruction, combined forecasting model generation, and optimization strategy. The overall framework of the proposed model is shown in Fig. 4. The detailed implementation steps are as follows.

Step 1: Data collection: Collecting wind power data and day-ahead numerical weather prediction (NWP) in Ningxia and Jilin, China.

Step 2: Selection of key meteorological factors: Calculating the correlation between 22 typical NWP meteorological variables (such as wind speed, wind direction, temperature, and pressure, etc.) and wind power. If the correlation value is higher than the set threshold, the best input meteorological variable is selected.

Step 3: Data decomposition and reconstruction: VMD is used to decompose the wind power sequence into several modes, and calculate the weight permutation entropy value of each mode, and then reconstruct the decomposed data into high, medium, low complexity sequence according to the principle of data similarity degree.

Step 4: Input variable feature extraction: The features of key factors and reconstruction modes can be extracted by CNN that using convolution, kernel, and pooling operations, thereby the shortcomings of traditional neural networks that lack useful information extraction can be overcome.

Step 5: Deep learning prediction model: The variables obtained by CNN are used as the LSTM prediction model input. In this step, different optimizers are used to optimize the prediction model to obtain prediction values based on different optimizers.

Step 6: Aggregate technique: Using the aggregation technique to re-add the prediction results obtained from different optimizers, and finally obtain the day-ahead wind power forecasting.

Step 7: Forecasting performance evaluation: Evaluating the performance of each prediction model by using multiple error evaluation metrics. The prediction results show that the proposed method outperforms other benchmark models in terms of prediction accuracy and improved performance percentage.

In the implementation operation, a set of training data can be regarded as a time window, and the proposed model moves the data window regularly (e.g., one day). This operating mechanism can ensure that the proposed model has high practicability in practice.

3. Data analysis and performance criteria

In this section, the performance of the proposed prediction

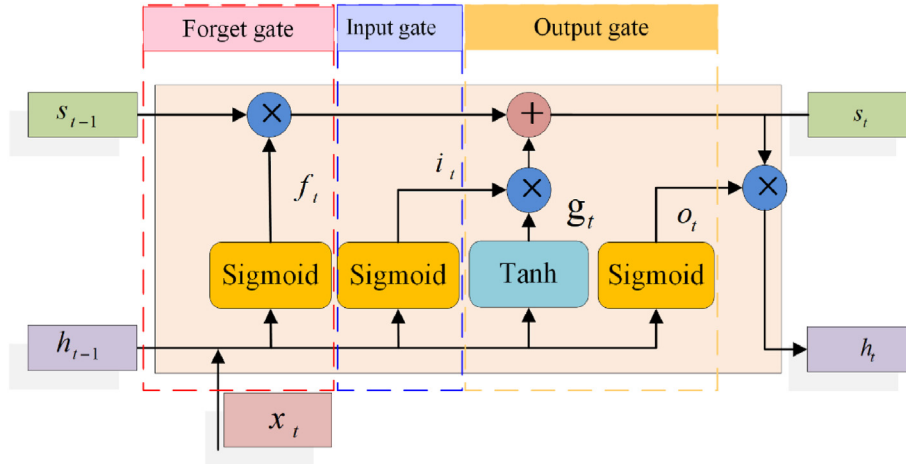


Fig. 3. The internal structure of the LSTM unit.

framework will be evaluated by four wind farms data in Jilin and Ningxia, China, and the selecting wind power data in different time and space is more representative. All experiments are running on Windows 10 with Intel Core i7-9700 CPU, 3.00 GHz, 64-bit, and 16 GB RAM.

3.1. Data collection and description

To evaluate the effectiveness of the proposed method, four experiments are carried out with the data from typical wind farms in Jilin and Ningxia wind power bases. The location distribution of typical wind farms is shown in Fig. 5. Taking into account the influence of seasonal factors on wind power forecasting, the typical wind farms A-D of the wind power base are selected. The selected wind farm A and wind farm B are located in Jilin Province of China, and the installed capacity of the wind farm are 200 MW and 198 MW, respectively. The selected wind farm C and wind farm D are located in the Ningxia Hui Autonomous Region of China, and the installed capacity of the wind farm C and wind farm D are 297 MW and 150 MW, respectively. The total number of power values of each wind farm is 2880, covering 30 days from June 1, 2018 at 02:00, to July 1, 2018 at 02:00, with a time horizon of 15 min. The detailed statistical description of the power data is shown in Table 1. The four wind farms are selected can characterize the fluctuation of wind power output in different in time and space, and can verify the effectiveness of the proposed method.

3.2. Performance criteria

In this study, normalized root mean square error (NRMSE), normalized mean absolute error (NMAE), accuracy rate (A_R) and correlation coefficient (R) are used to evaluate the superiority of the proposed model [37–39], and two improvement percentage metrics are used to evaluate the accuracy of the forecasting model. The error metrics are as follows:

$$NMAE = \frac{1}{n} \sum_{i=1}^n \left(\frac{|P_{i,t}^{real} - P_{i,t}^{pre}|}{Cap} \right) \quad (17)$$

$$NRMSE = \frac{\sqrt{\sum_{i=1}^n (P_{i,t}^{real} - P_{i,t}^{pre})^2}}{Cap \cdot \sqrt{n}} \quad (18)$$

$$A_R = \left(1 - \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{P_{i,t}^{real} - P_{i,t}^{pre}}{Cap} \right)^2} \right) \quad (19)$$

$$R = \frac{\sum_{i=1}^n [(P_{i,t}^{real} - \bar{P}_{i,t}^{real})(P_{i,t}^{pre} - \bar{P}_{i,t}^{pre})]}{\sqrt{\sum_{i=1}^n (P_{i,t}^{real} - \bar{P}_{i,t}^{real})^2} \sqrt{\sum_{i=1}^n (P_{i,t}^{pre} - \bar{P}_{i,t}^{pre})^2}} \times 100\% \quad (20)$$

where $P_{i,t}^{pre}$ is the prediction value at time t , $P_{i,t}^{real}$ is the actual wind power at time t , $\bar{P}_{i,t}^{real}$ is the average actual value at time t , $\bar{P}_{i,t}^{pre}$ is the average prediction value at time t , W_{cap} is the installed capacity of a wind farm, which are set 200.5 MW, 198 MW, 297 MW, and 150 MW, respectively.

To quantitatively compare the performance of each prediction model, the improvement percentages metrics are introduced [38,40,41]. Δ_{NMAE} and Δ_{NRMSE} are the improvement percentages of NMAE and NRMSE, respectively. The improvement percentages metrics can be described as:

$$\begin{cases} \Delta_{NMAE} = \frac{NMAE_{\text{benchmark model}} - NMAE_{\text{proposed model}}}{NMAE_{\text{benchmark model}}} \times 100\% \\ \Delta_{NRMSE} = \frac{NRMSE_{\text{benchmark model}} - NRMSE_{\text{proposed model}}}{NRMSE_{\text{benchmark model}}} \times 100\% \end{cases} \quad (21)$$

4. Results and analysis

4.1. Selection of key meteorological variables

The 22 NWP variables related to wind power are shown in Fig. 6 shows, and the detailed NWP variables are shown in Table 2. It can be seen that the correlation between meteorological factors and

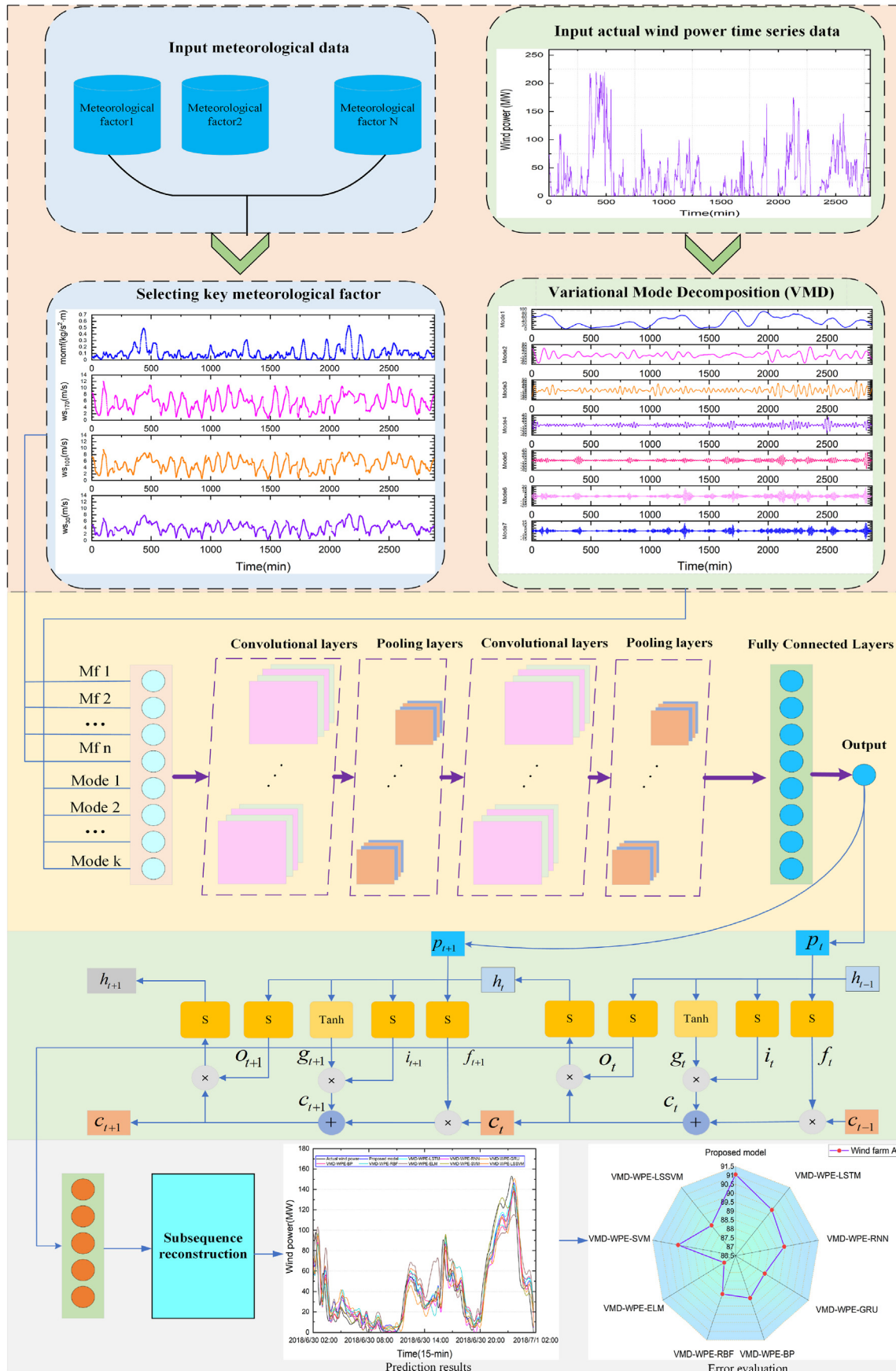


Fig. 4. The details overall framework of the proposed model.

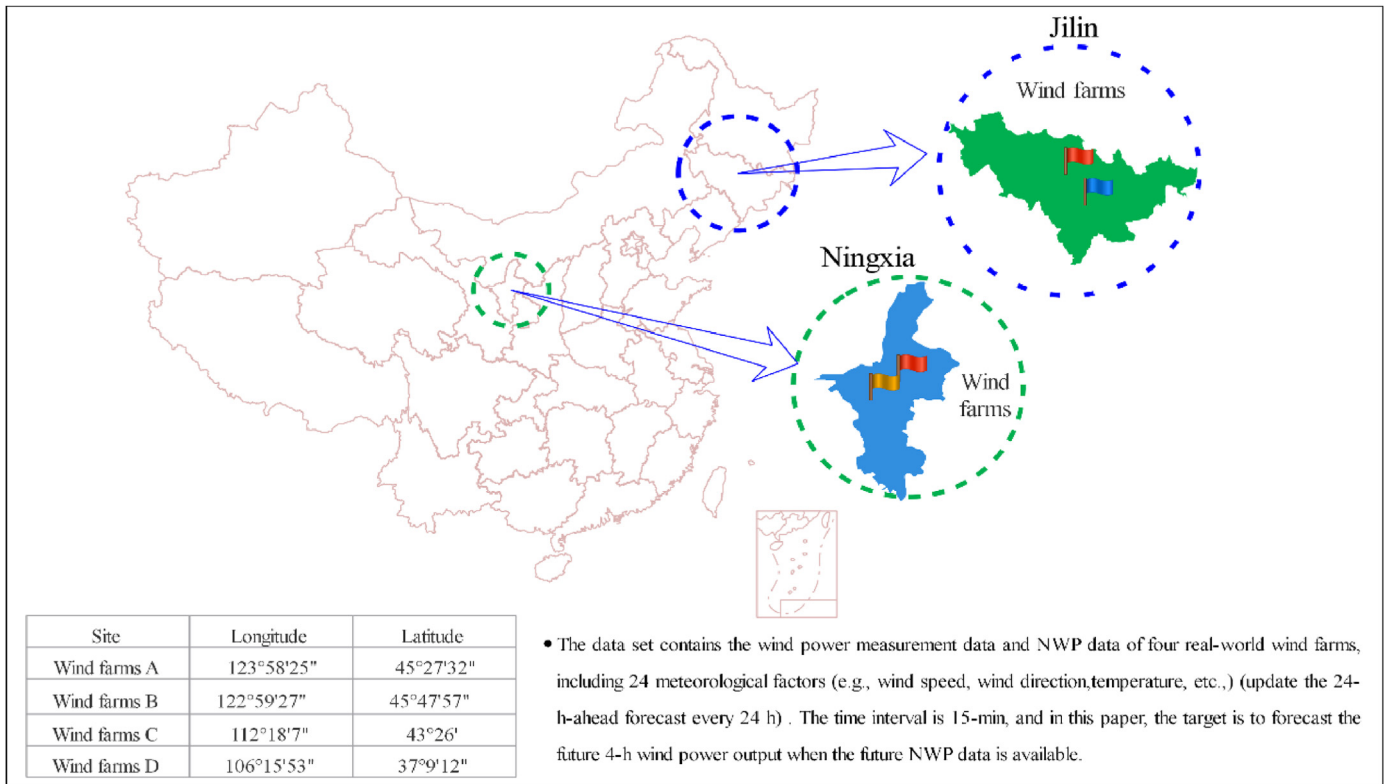


Fig. 5. Locations of the four wind farms in China.

Table 1
Statistics of the wind power data.

Site	Mean (MW)	Std.dev. (MW)	Max. (MW)	Min. (MW)	Capacity (MW)
Wind farms A	50.65	14.77	184.90	0	200
Wind farms B	58.95	12.29	188.30	0	198
Wind farms C	53.65	6.85	258.50	0	297
Wind farms D	41.44	31.70	148.47	0	150

wind power is different. Therefore, it is necessary to select meteorological factors before establishing a wind power prediction model.

Fig. 7 shows the correlation coefficients between different meteorological variables and wind power, the correlation threshold is set 0.5, and the threshold is greater than 0.5, the variable is selected, and vice versa. According to the correlation coefficient threshold value, the best variable can be selected. For wind farm A-C, the meteorological variables (e.g., momf, ws_{170} , ws_{100} , ws_{30} , and ws_{10}) are used as the best input variables. Especially, for wind farm D, in addition to the above five kinds of best meteorological variables, the correlation coefficient between variable wd_{170} and wind power also exceeds 0.5, so the wd_{170} is also selected as the best input variable. The selected best variable and the reconstruction sequence are used as sample input to train the prediction model.

4.2. Wind power decomposition and reconstruction analysis

In the data decomposition stage, the original wind power is decomposed by the VMD algorithm, and the WPE is used to characterize the complexity of the decomposed wind power. The choice of K value affects the occurrence of modal aliasing. Therefore, an appropriate K value must be selected. In the experiment, $K = 6$, $K = 7$, $K = 8$, and $K = 9$ are used to decompose the wind power,

respectively, the iteration curves of the modal center frequencies of the different K values are as shown in Fig. 8. In Fig. 8, it can be seen that when $K = 9$, the iterated curves of the Mode 8 is almost coincided with the Mode 9. In other words, modal aliasing appears. Therefore, the mode number is taken as 8.

Taking wind farm B as an example, the VMD algorithm decomposes the original wind power into 8 modes through the center frequency method. Fig. 9 shows the decomposition result of wind power, and the WPE values of all modes are shown in Table 3. It can be seen that the WPE values of each mode are increased with the mode frequency from low to high, which verified the effectiveness of WPE. The larger the WPE value, the higher the complexity of the mode. To improve the efficiency of the prediction model, the modes with similar WPE values are merged to construct a new mode. The new construction mode is shown in Fig. 10. It can be found that the number of modes has been reduced, which indicated that the complexity of the model input variable is reduced.

4.3. Performance analyses based on different optimization strategy

In the field of deep learning, the performance of the forecasting model depends on the optimizer. This section presents the effects of different optimizers on improving prediction accuracy. Four optimizers (e.g., Momentum, RMSProp, AdaGrad, Adam) are used to

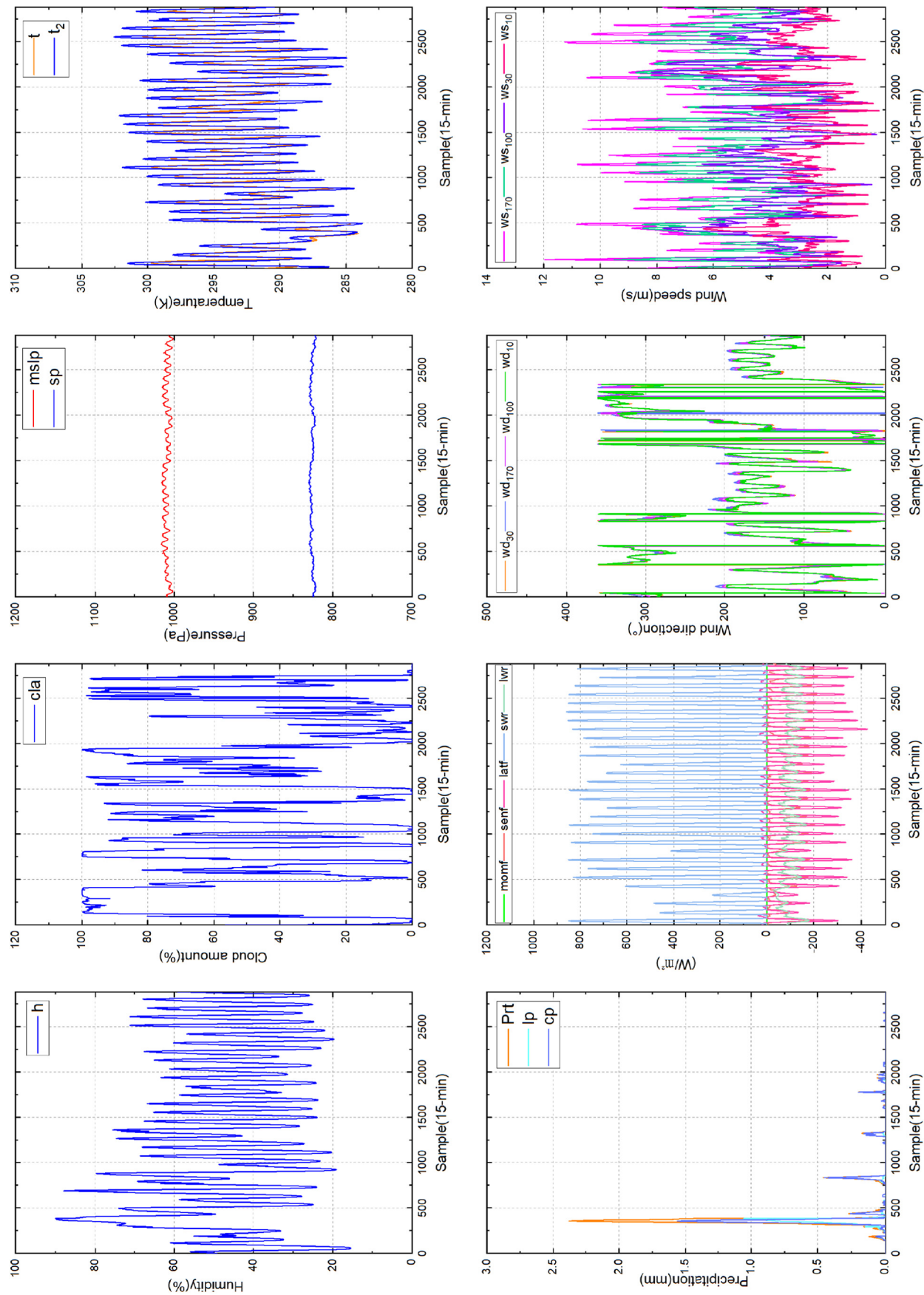


Fig. 6. The NWP variables related to wind power.

Table 2
Description of NWP variables.

NWP number	Variables	Description	Unit	NWP number	Variables	Description	Unit
1	ws ₁₇₀	Wind speed at 170 m	m/s	12	T	Temperature	K
2	ws ₁₀₀	Wind speed at 100 m	m/s	13	H	Humidity	%
3	ws ₃₀	Wind speed at 30 m	m/s	14	momf	Momentum flux	W/m ²
4	ws ₁₀	Wind speed at 10 m	m/s	15	cp	Convective precipitation	mm
5	wd ₁₇₀	Wind direction at 170 m	°	16	lp	Large-scale precipitation	mm
6	wd ₁₀₀	Wind direction at 100 m	°	17	lwr	Long wave radiation	W/m ²
7	wd ₃₀	Wind direction at 30 m	°	18	latf	Latent heat flux	W/m ²
8	wd ₁₀	Wind direction at 10 m	°	19	senf	Sensible heat flux	W/m ²
9	mslp	Sea level pressure	Pa	20	swr	Shortwave radiation	W/m ²
10	sp	Surface pressure	Pa	21	prt	Precipitation amount	mm
11	t ₂	Temperature at 2 m	K	22	cla	Cloud amount	%

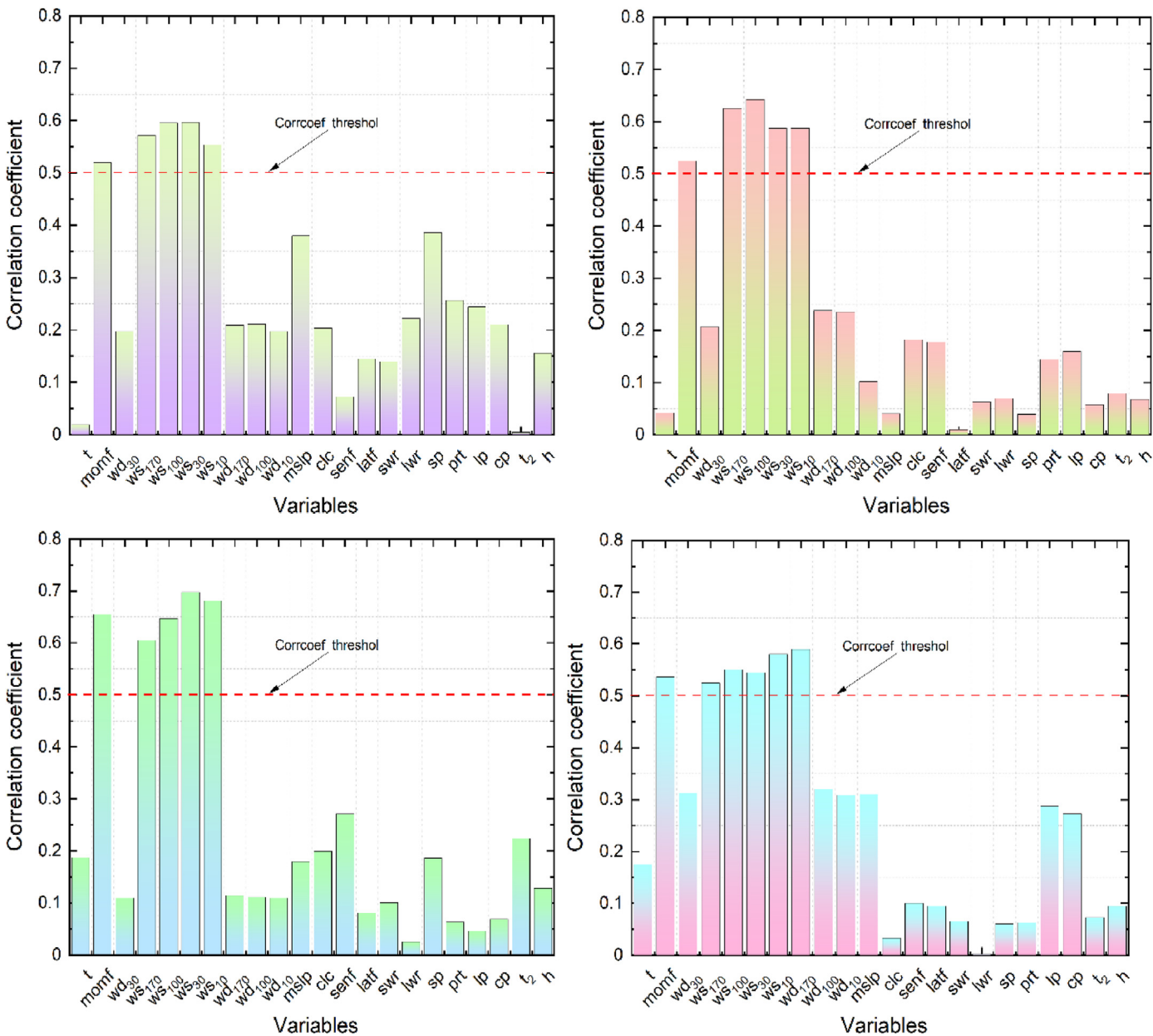


Fig. 7. The correlation coefficients between meteorological variables and wind power.

optimize the parameters of the CNN-LSTM forecasting model, respectively. The new mode I, new mode II, new mode III, and new

model IV are used for testing the forecasting model. Table 4 shows the prediction results of different optimizers.

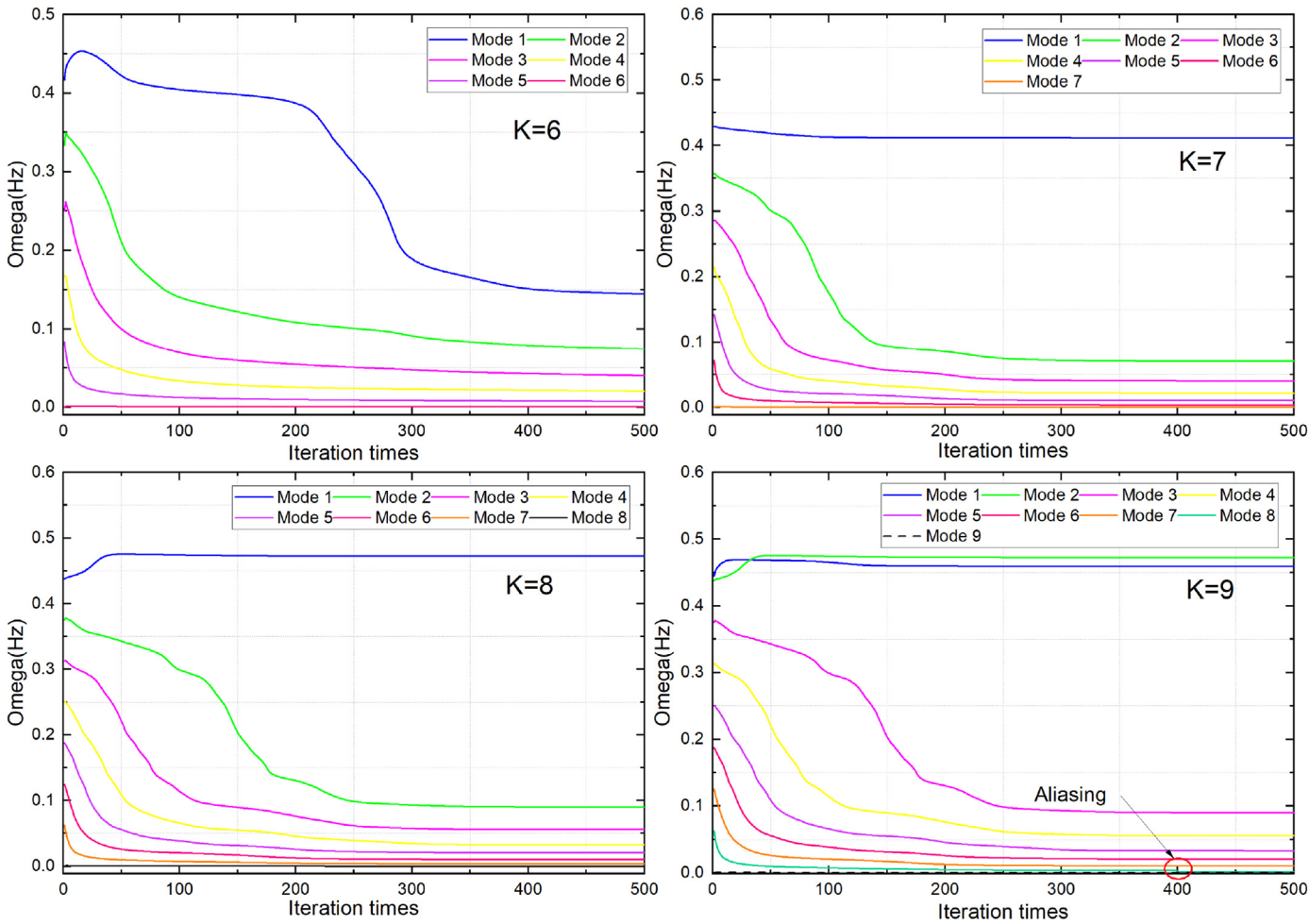


Fig. 8. Center frequency iteration curve.

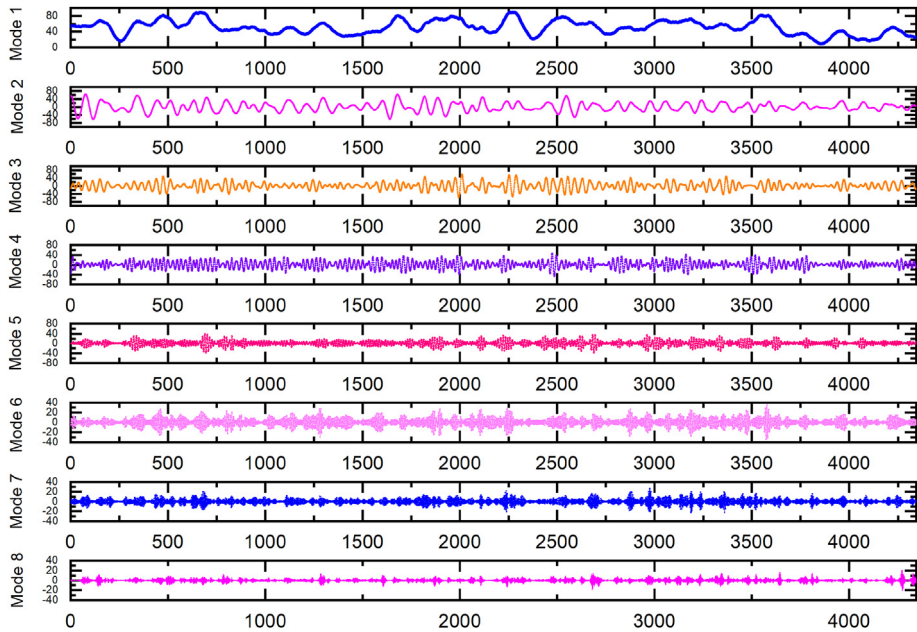


Fig. 9. Decomposition by VMD of wind power.

Table 3
The MPE values of all modes.

Original sequence	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6	Mode 7	Mode 8
MPE values	0.1501	0.1417	0.1349	0.1395	0.3263	0.4841	0.6032	0.6136
Reconstruction sequence	New model I				New model II		New model III	
Reconstruction value	0.5662				0.3263		0.4841	
							1.2168	

It can be seen that different optimizers have different performances in all test data. In the new mode I, the *NMAE* and *NRMSE* of the prediction model based on the Momentum optimizer are lower than other optimizers. In the new mode II subsequence, the prediction model based on RMSProp optimizer has lower *NMAE* and *NRMSE* than other optimizers. In the new mode III, the *NMAE*, *NRMSE*, and A_R of the prediction model based on the AdaGrad optimizer are 2.69, 3.63, and 0.8327, respectively, which are lower than other optimizers. In the new mode IV subsequence, the *NMAE*, *NRMSE*, and A_R of the prediction model based on Adam optimizer are 2.28, 3.27, and 0.8451, respectively, which are lower than other optimizers. In the above experiment, the forecasting model based on the AdaGrad optimizer requires more time to train the model due to the slow learning rate. Comprehensive analysis of the above experiments, different optimizers produce different forecasting results. Therefore, this paper selects different optimizers to optimize the forecasting model parameters according to the characteristic of the reconstruction mode data, which can improve forecasting accuracy.

4.4. The results of wind power forecasting

4.4.1. Forecasting results in different ahead hours

This section will present the results of wind power prediction, and three latest deep learning prediction models (e.g., CNN model, LSTM model, GRU model) and five popular prediction models (e.g., BP model, RBF model, ELM model, SVM model, and LSSVM model) are used as the benchmark model. Table 5 and Fig. 11 show the result of wind power forecasting in 1-h ahead, 6-h ahead, and 12-h ahead. It can be seen from Fig. 11 that the prediction performance of all prediction models is decreased as the prediction time horizon increases, especially, the prediction performance of the VMD-WPE-RBF model is significantly decreased, while the proposed model can track the actual power curve well, and the error curve is also relatively flat. Therefore, the error value is smaller than the other models. Compared with other benchmark models, a better WPF performance than other models is obtained. For example, for wind

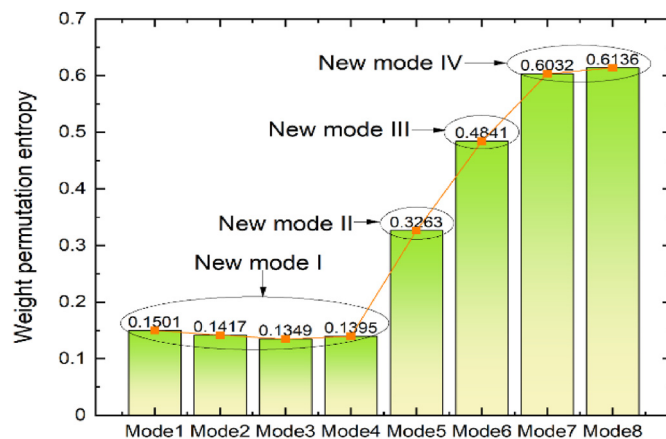


Fig. 10. The new construction mode by WPE of wind power.

Table 4
The prediction results of different optimizers.

Error metrics	Momentum optimizer			
	New model I	New model II	New model III	New model IV
<i>NMAE</i>	5.76	5.12	5.45	5.36
<i>NRMSE</i>	6.43	6.96	6.68	6.74
A_R	0.8712	0.8746	0.8798	0.8762
RMSProp optimizer				
	New model I	New model II	New model III	New model IV
<i>NMAE</i>	5.27	5.23	5.68	5.79
<i>NRMSE</i>	6.52	6.36	6.97	7.02
A_R	0.8779	0.8803	0.8761	0.8711
AdaGrad optimizer				
	New model I	New model II	New model III	New model IV
<i>NMAE</i>	5.43	5.48	5.32	5.76
<i>NRMSE</i>	6.64	6.68	6.57	6.96
A_R	0.8737	0.8832	0.8797	0.8703
Adam optimizer				
	New model I	New model II	New model III	New model IV
<i>NMAE</i>	5.82	5.73	5.95	5.31
<i>NRMSE</i>	6.53	6.47	6.75	6.36
A_R	0.8772	0.8763	0.8757	0.8718

farm A, the error evaluation metrics *NRMSE* in 1-h ahead, 6-h ahead, and 12-h ahead are 4.19, 5.00, and 7.33, respectively. It can be seen from Table 4 that for the test time horizons of 1-h ahead, 6-h ahead, and 12-h ahead, the proposed model outperforms other benchmark models.

4.4.2. Day-ahead wind power forecasting results

To illustrate the prediction results of each prediction method, Fig. 12 (a)–(d) shows the actual power and forecasting power curves of the four wind farms with the time horizon of day-ahead. It can be seen from Fig. 12 that all forecasting models track the trend of wind power and maintain the stability of prediction. Compared with other benchmark methods, the proposed model can better close the actual power curve, and it can be observed that the deviation between the forecasting value and the actual value is small. Although the VMD-WPE-ELM and VMD-WPE-RBF model can describe the overall trend of the wind power curve, they cannot track the peak and valley of wind power. Comparatively speaking, compared with other benchmark methods, the proposed method can better track wind power peak and valley. Considering the above discussion, it can be concluded that the proposed model combines the advantages of meteorological knowledge and data feature extraction, not only can reflect the overall characteristics of the wind power but also the distribution characteristics of the forecasting results are closer to the actual power.

The error metrics (e.g., *NMAE* and *NRMSE*) are used to evaluate the performance between the actual value and forecasting value. Table 6 shows the error metrics of the four wind farms in different forecasting models, and the forecasting time horizon is day-ahead. In the four wind power prediction experiments, the *NRMSE* metrics of the proposed method are smaller than other benchmark models, which indicates that the small errors and contribute to the *RMSE*. Compared with the prediction models based on deep learning (e.g., VMD-WPE-LSTM, VMD-WPE-RNN, VMD-WPE-GRU), the *NMAE* and *NRMSE* values of the proposed method are not significantly

Table 5
Error values for the multiple ahead time.

Sites	Models	NRMSE (%)			NMAE (%)			R		
		1-h	6-h	12-h	1-h	6-h	12-h	1-h	6-h	12-h
Wind farm A	Proposed model	4.19	5.00	7.33	3.72	4.21	5.14	0.9104	0.8821	0.8643
	VMD-WPE-LSTM	5.01	6.95	7.50	4.54	5.03	5.88	0.9052	0.8667	0.8371
	VMD-WPE-RNN	5.34	6.01	7.57	4.97	5.35	6.98	0.9011	0.8785	0.8615
	VMD-WPE-GRU	5.43	6.08	7.42	4.83	5.68	6.23	0.9096	0.8783	0.8597
	VMD-WPE-BP	8.94	11.79	13.57	6.06	8.85	10.66	0.8692	0.8669	0.8640
	VMD-WPE-RBF	9.48	13.33	15.29	6.51	9.95	12.64	0.8771	0.8744	0.8704
	VMD-WPE-ELM	11.76	12.98	14.53	8.36	9.71	11.50	0.8824	0.8702	0.8547
	VMD-WPE-SVM	9.19	12.15	13.85	5.90	8.07	9.54	0.8726	0.8628	0.8655
	VMD-WPE-LSSVM	9.04	12.17	14.03	6.24	9.16	11.21	0.8754	0.8699	0.8735
Wind farm B	Proposed model	4.96	5.51	7.96	3.51	4.82	5.29	0.9187	0.8853	0.8592
	VMD-WPE-LSTM	5.75	6.84	7.55	4.16	5.86	5.51	0.9098	0.9104	0.8895
	VMD-WPE-RNN	5.93	6.06	7.14	4.46	5.57	6.98	0.9005	0.8876	0.8663
	VMD-WPE-GRU	5.75	6.56	7.02	4.72	5.43	6.18	0.9021	0.8886	0.8696
	VMD-WPE-BP	11.12	14.73	15.80	7.39	10.92	13.22	0.8888	0.8527	0.8320
	VMD-WPE-RBF	11.88	13.06	14.77	8.23	11.01	13.75	0.8818	0.8394	0.8123
	VMD-WPE-ELM	12.98	13.49	15.79	9.51	10.70	12.41	0.8702	0.8351	0.8221
	VMD-WPE-SVM	11.45	13.07	15.01	7.28	7.81	9.81	0.8855	0.8493	0.8331
	VMD-WPE-LSSVM	11.32	14.83	16.80	7.77	10.97	13.17	0.8868	0.8517	0.8320
Wind farm C	Proposed model	5.03	6.51	7.24	4.89	5.57	6.42	0.9121	0.8886	0.8765
	VMD-WPE-LSTM	6.28	6.79	7.42	4.16	5.86	5.51	0.9098	0.8804	0.8795
	VMD-WPE-RNN	6.95	7.24	8.37	5.46	5.44	6.84	0.8905	0.8776	0.8663
	VMD-WPE-GRU	6.79	7.14	8.04	5.03	6.57	6.42	0.8821	0.8786	0.8696
	VMD-WPE-BP	7.76	11.14	13.08	4.74	7.88	9.33	0.9224	0.8886	0.8692
	VMD-WPE-RBF	11.88	13.06	14.77	8.23	11.01	13.64	0.8812	0.8394	0.8123
	VMD-WPE-ELM	11.46	12.65	13.70	8.15	8.99	10.11	0.8854	0.8735	0.8630
	VMD-WPE-SVM	7.72	11.22	12.47	4.64	6.89	7.53	0.9228	0.8878	0.8732
	VMD-WPE-LSSVM	7.79	11.14	13.04	5.03	7.57	9.42	0.9221	0.8886	0.8696
Wind farm D	Proposed model	5.13	6.47	7.08	4.75	5.42	6.93	0.9187	0.8953	0.8892
	VMD-WPE-LSTM	6.02	6.96	7.05	5.13	5.57	6.37	0.8998	0.8704	0.8795
	VMD-WPE-RNN	6.39	6.24	7.02	5.86	5.37	6.49	0.8861	0.8796	0.8698
	VMD-WPE-GRU	6.87	6.54	7.10	5.54	5.57	6.94	0.8813	0.8746	0.8690
	VMD-WPE-BP	7.90	11.63	14.24	5.55	8.70	11.14	0.9210	0.8837	0.8576
	VMD-WPE-RBF	9.48	13.33	14.27	6.51	9.95	12.46	0.9052	0.8667	0.8371
	VMD-WPE-ELM	11.93	13.62	15.61	8.54	11.56	11.88	0.8807	0.8238	0.8439
	VMD-WPE-SVM	8.09	12.00	13.01	5.41	8.20	10.37	0.9191	0.8800	0.8499
	VMD-WPE-LSSVM	7.78	11.54	14.10	5.54	8.57	10.94	0.9213	0.8846	0.8590

reduced. Compared with VMD-WPE-BP, VMD-WPE-RBF, VMD-WPE-ELM, VMD-WPE-SVM, and VMD-WPE-LSSVM, the proposed method can achieve the smallest errors. It can be seen that the performance of the deep model is better than the traditional neural network (e.g., BP, RBF, ELM) and machine learning model (e.g., SVM and LSSVM), while the proposed model outperforms all other benchmark models.

In terms of relative error distribution, Fig. 13 shows the fluctuations of all prediction models errors. It can be seen that the error curve of the proposed method always oscillates up and down on the zero axes, and there is no maximum value, and the error curve is smoother than other benchmark models. This also indicates that at the level of wind power peak, the proposed method shows better performance than other benchmark methods, which also shows that the prediction stability of the model can be obtained.

To visualize the performance of the proposed method and other benchmark models, the prediction accuracy A_R of the four wind farms is used to evaluate the performance of each prediction model. Fig. 14 shows the forecasting accuracy results in different models. It can be found that in terms of wind farm A, B, and D, the VMD-WPE-ELM model is the closest to the small circle than other models, indicating that the VMD-WPE-ELM method has the worst forecasting accuracy. For wind farm C, the VMD-WPE-LSSVM model is close to the small circle, which indicates that the method obtains the worst prediction accuracy. The detailed data results are shown in Table 7. In all wind farm experiments, the forecasting accuracy of the proposed method is closest to the great circle, which indicates that the accuracy of the proposed method is superior to other

benchmark methods.

5. Discussion

5.1. Discussion of the improvement percentages forecasting performance

In this section, comprehensively test experiments are conducted to evaluate the effectiveness of the proposed model, taking day-ahead wind power forecasting as an example, Table 8 shows the results of Δ_{NMAE} and Δ_{NRMSE} in different methods for four wind farms. In the cases of wind farm A, B, C, and D, compared with the VMD-WPE-LSTM model, the Δ_{NRMSE} of the proposed model is reduced by 17.92%, 19.71%, 23.30%, 9.69%, respectively; compared with the VMD-WPE-RNN model, the Δ_{NRMSE} of the proposed model is reduced by 21.53%, 20.94%, 23.51%, 15.44%, respectively; compared with the VMD-WPE-GRU model, the Δ_{NRMSE} of the proposed model is reduced by 12.57%, 26.65%, 16.72%, 19.48%, respectively; Compared with other 5 benchmark forecasting models, the Δ_{NMAE} and Δ_{NRMSE} are also still decreases significantly. It indicates that the proposed model can obtain relatively reliable forecasting value. It further shows that the proposed prediction framework has contributed to the improvement of accuracy.

5.2. Discussion of time efficiency

Computational efficiency is an important indicator to evaluate the performance of the model, the computational time of the

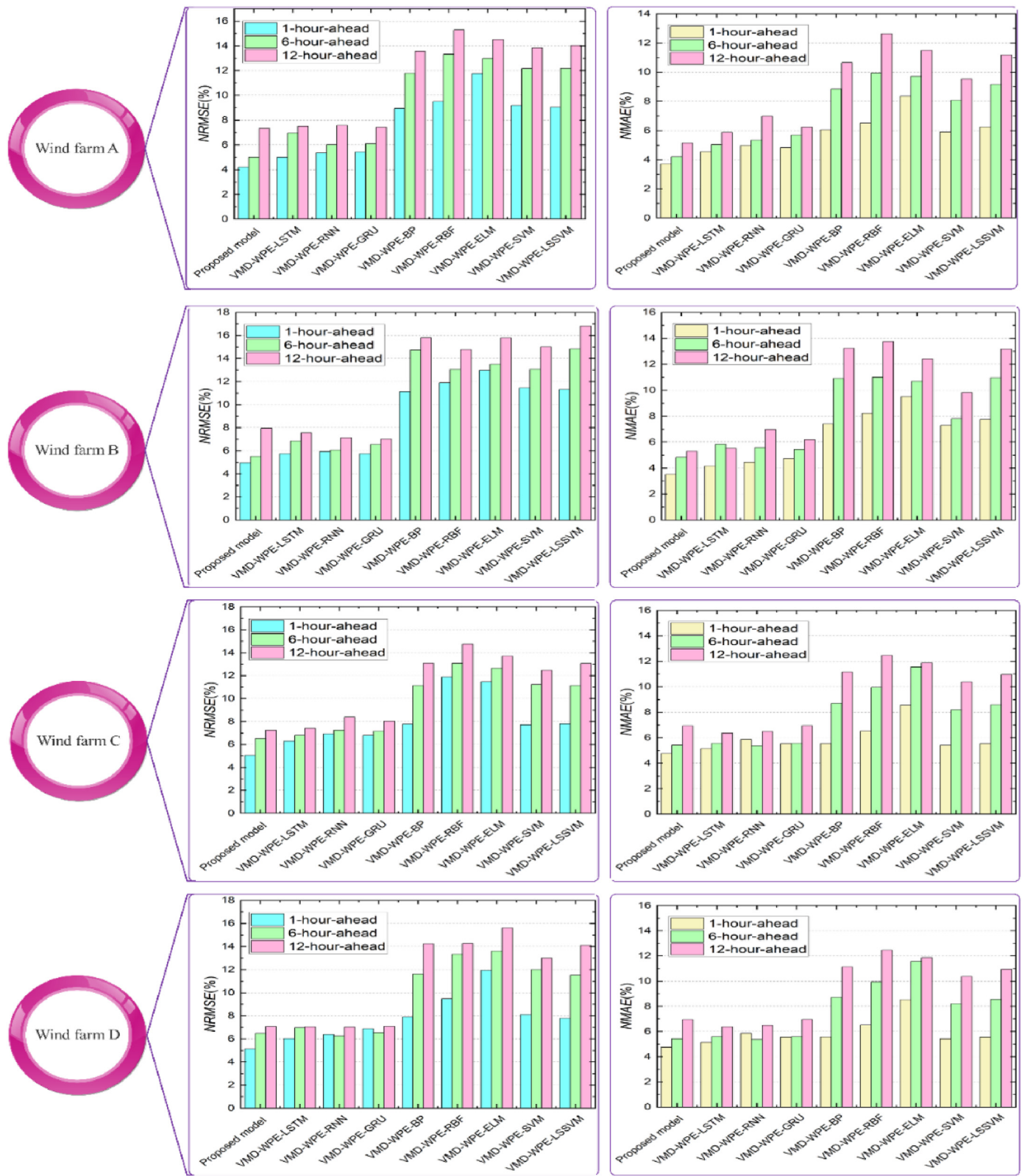


Fig. 11. The trend of error with multiple ahead time for different models.

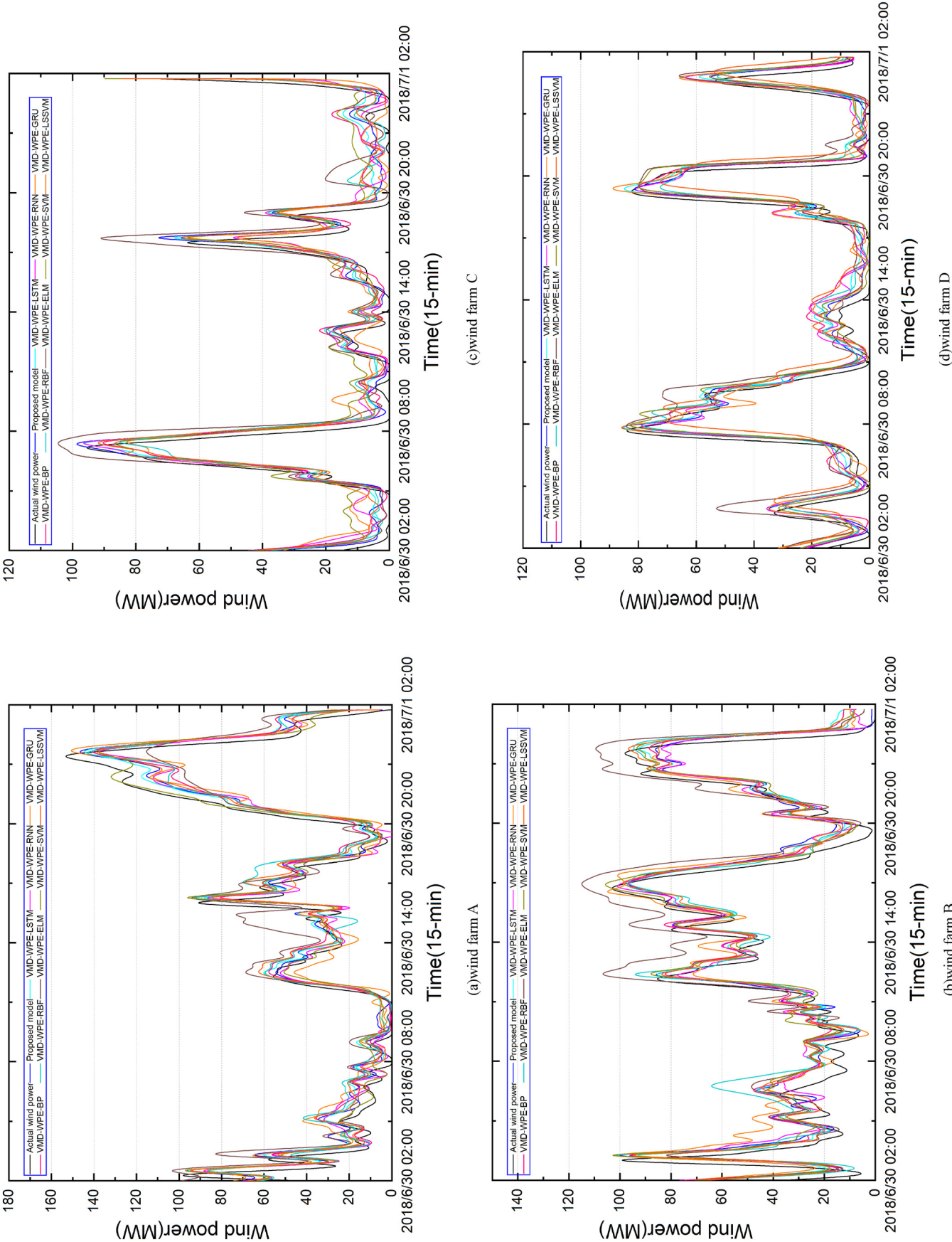


Fig. 12. Results of wind power prediction in different methods.

Table 6
Comparison of wind power prediction.

Methods	Wind farm A		Wind farm B		Wind farm C		Wind farm D	
	<i>NRMSE</i> (%)	<i>NMAE</i> (%)	<i>NRMSE</i> (%)	<i>NMAE</i> (%)	<i>NRMSE</i> (%)	<i>NMAE</i> (%)	<i>NRMSE</i> (%)	<i>NMAE</i> (%)
Proposed model	7.51	5.78	7.21	6.02	5.53	4.17	6.24	4.77
VMD-WPE-LSTM	9.15	7.42	8.98	6.39	7.21	5.11	6.91	4.94
VMD-WPE-RNN	9.57	7.93	9.12	7.58	7.23	5.18	7.38	5.61
VMD-WPE-GRU	8.59	7.88	8.09	7.34	6.64	5.33	7.75	5.72
VMD-WPE-BP	10.98	7.87	9.83	7.27	6.18	4.90	8.07	5.55
VMD-WPE-RBF	11.22	8.16	10.84	7.97	6.14	4.89	8.11	5.56
VMD-WPE-ELM	12.72	9.45	12.48	9.63	7.67	4.53	9.54	6.65
VMD-WPE-SVM	10.04	7.64	10.02	7.21	6.59	5.68	8.48	5.17
VMD-WPE-LSSVM	10.48	8.17	11.32	8.42	7.72	5.27	9.17	6.41

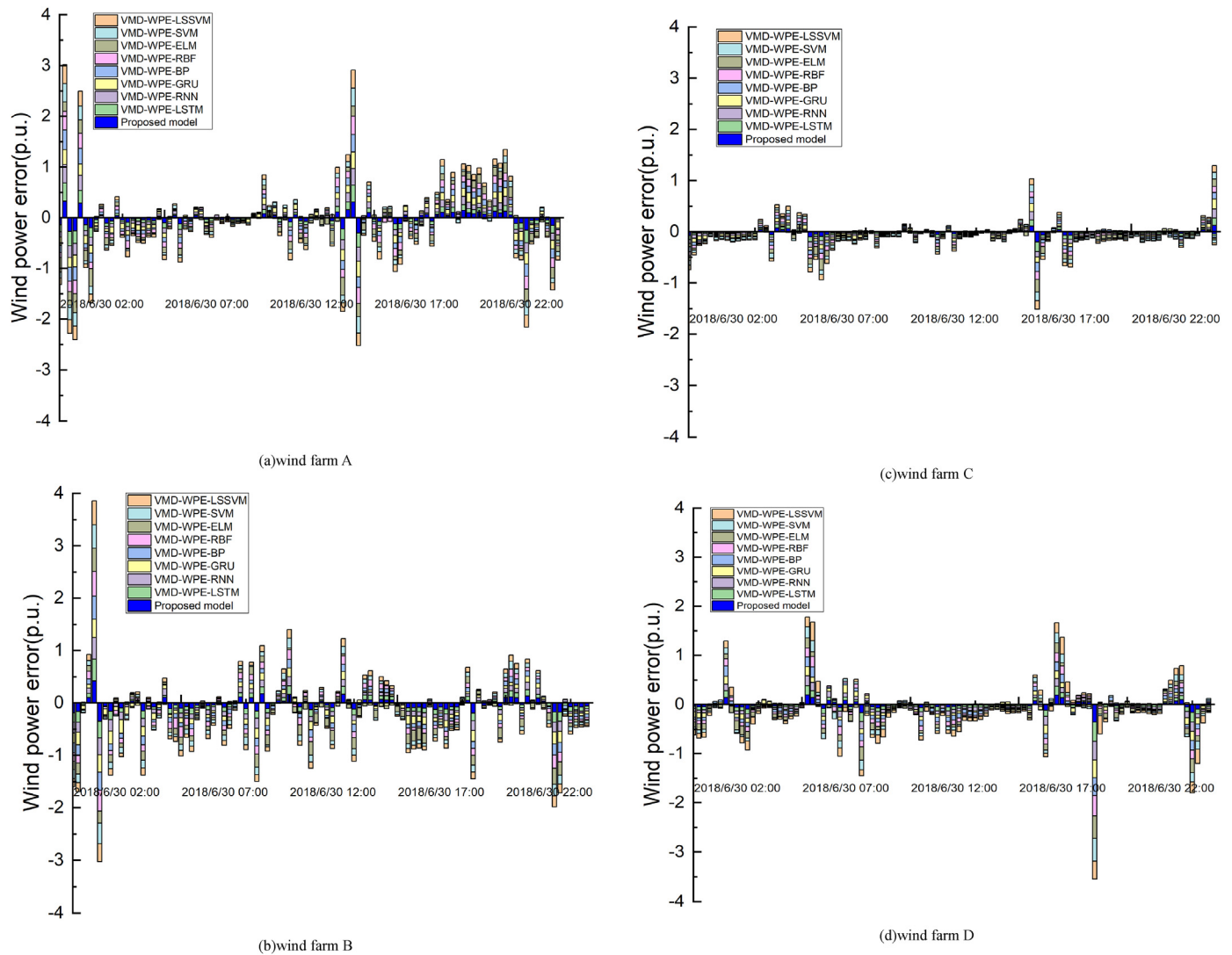


Fig. 13. The fluctuations of all prediction models errors.

proposed model and benchmark model are summarized in Table 9. It can be seen that the computation time of the proposed model in day-ahead wind power forecasting are 13.36s, 14.57s, 15.32s, and 16.21s for wind farm A, B, C, and D, respectively. It can be seen from Fig. 15 that the VMD-WPE-RBF model takes more time to forecast wind power, while the proposed method requires the least time compared to other benchmark models. The detailed information is shown in Table 9. It can be concluded that the computational

efficiency of the proposed model is more advantageous. Overall, this is acceptable in a practical wind power system.

6. Conclusion and future works

Day-ahead wind power forecasting plays a dominant role in smart grids. To improve the prediction accuracy and model reliability, a novel combined prediction framework is proposed, which

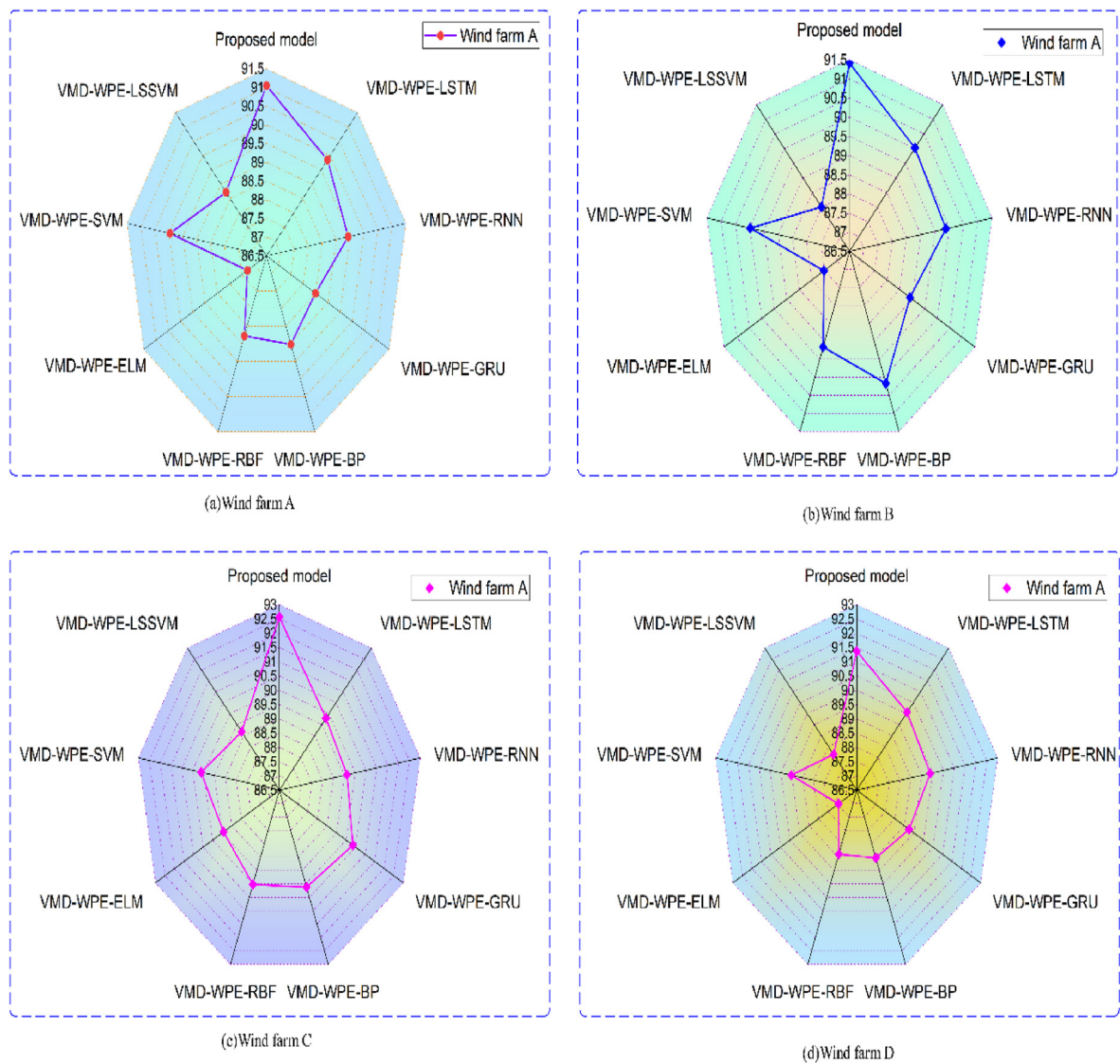


Fig. 14. Results of wind power prediction in different methods.

Table 7
Forecasting accuracy results for each forecasting method.

Methods	Wind farm A	Wind farm B	Wind farm C	Wind farm D
	A_R (%)	A_R (%)	A_R (%)	A_R (%)
Proposed model	91.05	91.42	92.57	91.37
VMD-WPE-LSTM	89.85	90.02	89.79	90.06
VMD-WPE-RNN	89.43	89.88	89.62	89.89
VMD-WPE-GRU	88.50	88.91	90.36	89.25
VMD-WPE-BP	89.02	90.17	90.12	89.03
VMD-WPE-RBF	88.78	89.16	90.02	88.89
VMD-WPE-ELM	87.28	87.52	89.43	87.46
VMD-WPE-SVM	89.96	89.98	90.11	89.52
VMD-WPE-LSSVM	88.72	88.01	89.17	88.14

includes key meteorological factors selection, data decomposition and reconstruction, combined forecasting model generation, and optimization strategy. In the data selection stage, five meteorological factors with the strongest correlation with wind power are

selected as the input of forecasting model. In the data decomposition and reconstruction stage, multiple frequency-domain modes can be obtained via VMD-WPE. In the combined forecasting model generation stage, a forecasting model based CNN-LSTM is developed to forecast day-ahead wind power. In the optimization strategy selection stage, different optimizers are used to optimize the prediction model to obtain reliable prediction results. To analyze the forecasting performance at different time horizons, such as 1-h ahead, 6-h ahead, 12-h ahead and 24-h ahead, four wind farms in China are used to test the proposed prediction method and compared with eight classic benchmark prediction methods. The results show that the proposed method outperforms other benchmarks taking into account all evaluation performance criteria.

In future work, different data feature extraction methods should be applied to obtain useful weather information. Besides, the heuristic optimization algorithm can try to optimize the parameters of the prediction model to improve the prediction

Table 8
The results of Δ_{NMAE} and Δ_{NRMSE} in different methods for four wind farms.

Prediction models	Indexes	Wind farm A	Wind farm B	Wind farm C	Wind farm D
VMD-WPE-LSTM vs. Proposed model	Δ_{NMAE} (%)	22.10	5.79	18.40	3.44
	Δ_{NRMSE} (%)	17.92	19.71	23.30	9.69
VMD-WPE-RNN vs. Proposed model	Δ_{NMAE} (%)	27.11	20.58	19.50	14.97
	Δ_{NRMSE} (%)	21.53	20.94	23.51	15.44
VMD-WPE-GRU vs. Proposed model	Δ_{NMAE} (%)	26.65	17.98	21.76	16.61
	Δ_{NRMSE} (%)	12.57	26.65	16.72	19.48
VMD-WPE-BP vs. Proposed model	Δ_{NMAE} (%)	26.56	17.19	14.90	14.05
	Δ_{NRMSE} (%)	31.60	26.65	10.52	22.68
VMD-WPE-RBF vs. Proposed model	Δ_{NMAE} (%)	29.17	24.47	14.72	14.21
	Δ_{NRMSE} (%)	33.07	33.49	9.34	23.06
VMD-WPE-ELM vs. Proposed model	Δ_{NMAE} (%)	38.84	37.49	7.95	28.27
	Δ_{NRMSE} (%)	40.96	42.23	27.90	34.59
VMD-WPE-SVM vs. Proposed model	Δ_{NMAE} (%)	24.35	16.50	26.58	7.74
	Δ_{NRMSE} (%)	25.19	28.04	16.08	26.41
VMD-WPE-LSSVM vs. Proposed model	Δ_{NMAE} (%)	29.25	28.50	20.87	25.59
	Δ_{NRMSE} (%)	28.34	36.31	28.37	31.95

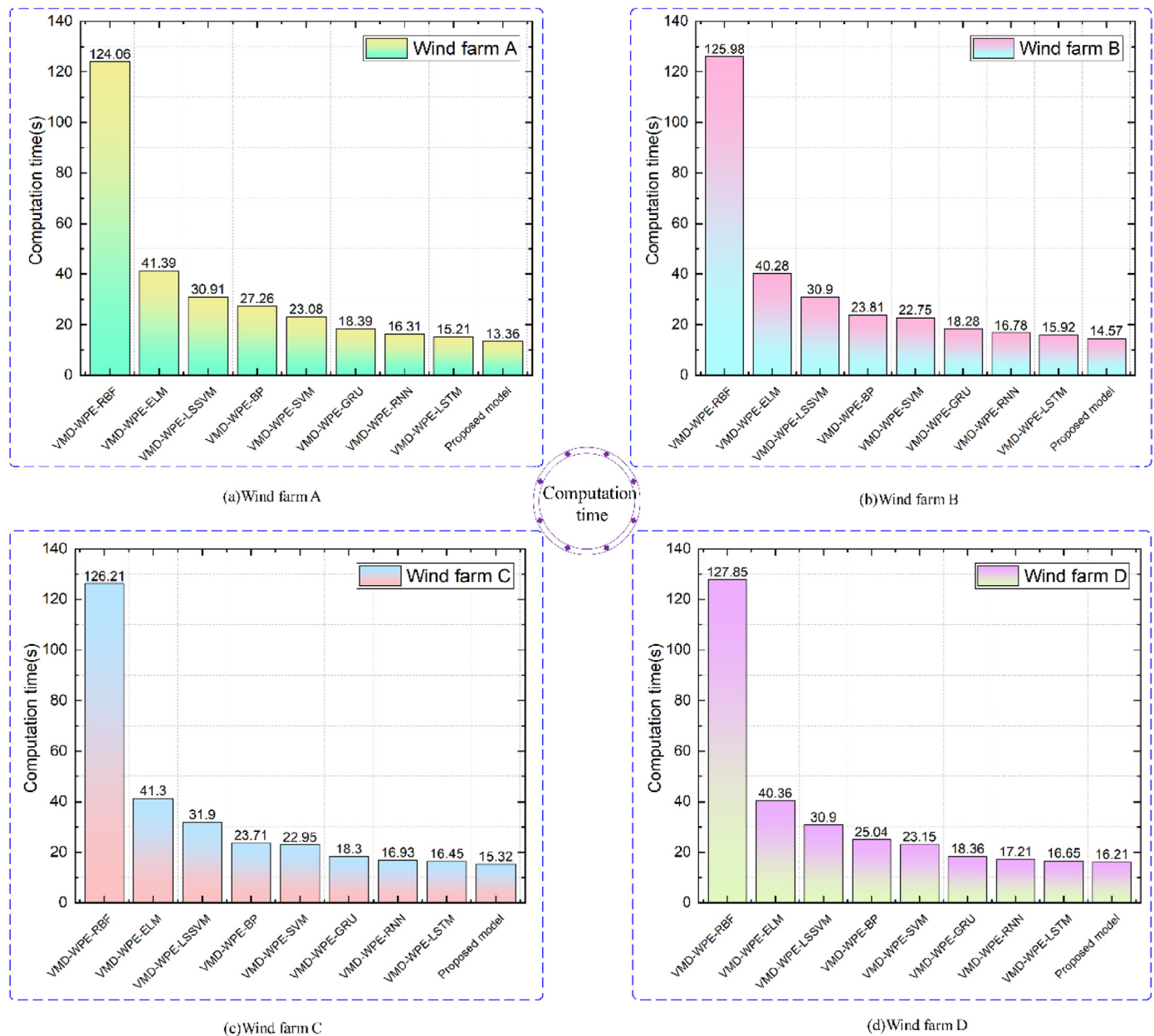


Fig. 15. Computation time for different WPF models.

Table 9

The computation time for each prediction model in four wind farm experiments.

Methods	Computation time (in Seconds)			
	Wind farm A	Wind farm B	Wind farm C	Wind farm D
VMD-WPE-RBF	124.06	125.98	126.21	127.85
VMD-WPE-ELM	41.39	40.28	41.3	40.36
VMD-WPE-LSSVM	30.91	30.9	31.9	30.9
VMD-WPE-BP	27.26	23.81	23.71	25.04
VMD-WPE-SVM	23.08	22.75	22.95	23.15
VMD-WPE-GRU	18.39	18.28	18.3	18.36
VMD-WPE-RNN	16.31	16.78	16.93	17.21
VMD-WPE-LSTM	15.21	15.92	16.45	16.65
Proposed model	13.36	14.57	15.32	16.21

performance. The proposed forecasting framework can also be applied to load demand, price, and other forecasting applications.

CRediT authorship contribution statement

Peng Lu: Conceptualization, Methodology, Software, Data curation, Visualization, Investigation, Writing – original draft. **Lin Ye:** Supervision, Validation, Resources, Writing – review & editing. **Ming Pei:** Investigation. **Yongning Zhao:** Writing – review & editing. **Binhua Dai:** Writing – review & editing. **Zhuo Li:** Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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